CLINICAL · NOSE & SINUSES

Allergic and non-allergic rhinitis

Rhinitis is the commonest complaint in the nose clinic: sneezing, a running nose, itch and a blocked nose. The history sorts allergic from non-allergic. Steroid sprays used properly control most patients, and turbinate surgery helps only when medicines have genuinely been tried. One-sided symptoms are never rhinitis until the nose has been examined.

AT A GLANCE

- Allergic or non-allergic by history.
- Topical steroid first, taught properly.
- Avoid decongestants beyond a week.
- One-sided symptoms aren't allergy.
- Immunotherapy is the only disease-modifying treatment; treat the nose to help asthma.

1 CHAPTER ONE Overview

Inflammation of the nasal lining.

1.1 Definition

Rhinitis is inflammation of the nasal mucosa, defined clinically by two or more of nasal obstruction, rhinorrhoea, sneezing and itching. *Rhinosinusitis* adds sinus involvement (see the sinusitis topics).

1.2 Classification

RHINITIS

- ALLERGIC **Seasonal or perennial** intermittent or persistent (ARIA)
- NON-ALLERGIC, NON-INFECTIVE **Vasomotor (idiopathic)** drug-induced, rhinitis medicamentosa, hormonal, NARES, gustatory, emotional, occupational

INFECTIVE **Acute and chronic rhinitis** atrophic rhinitis and specific infections (separate topics)

1.3 Why it matters

Allergic rhinitis affects up to a quarter of many populations, disturbs sleep, school and work, and travels with asthma: rhinitis patients are about four times more likely to develop asthma, and 20 to 30% of patients with rhinitis have asthma.

2 CHAPTER TWO Applied physiology

Why the nose runs and blocks.

2.1 The erectile mucosa

The turbinate mucosa contains venous sinusoids surrounded by smooth muscle that fill and empty them. Sympathetic stimulation empties them and opens the airway; parasympathetic stimulation fills them and drives the glands to secrete. Their alternating congestion from side to side is the *nasal cycle*.

2.2 Hyperreactivity

In both allergic and non-allergic rhinitis the mucosa over-reacts to ordinary stimuli: temperature change, dust, smoke, perfume, cold air.

3 CHAPTER THREE Allergic rhinitis

IgE in the nose.

3.1 Types

- **Seasonal:** symptoms in the season of the pollen the patient reacts to (tree, grass or weed pollens).

- **Perennial:** symptoms throughout the year, from house dust mite, moulds, cockroaches and animal dander.

3.2 Genetic predisposition

The chance of a child developing allergy is about 20% with one allergic parent and 47% with two.

3.3 Pathogenesis

3.3.1 Sensitisation

In predisposed people, inhaled allergens drive production of specific IgE, which binds by its Fc end to mast cells and basophils.

3.3.2 Re-exposure

Allergen bridges two IgE molecules on the cell surface; the mast cell degranulates, releasing preformed mediators (histamine) and newly made ones (leukotrienes, prostaglandins).

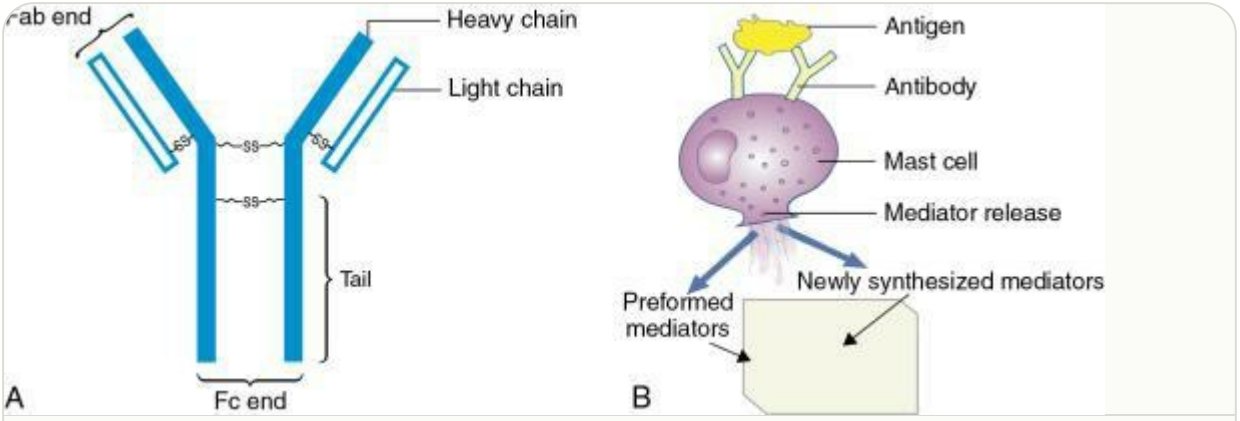


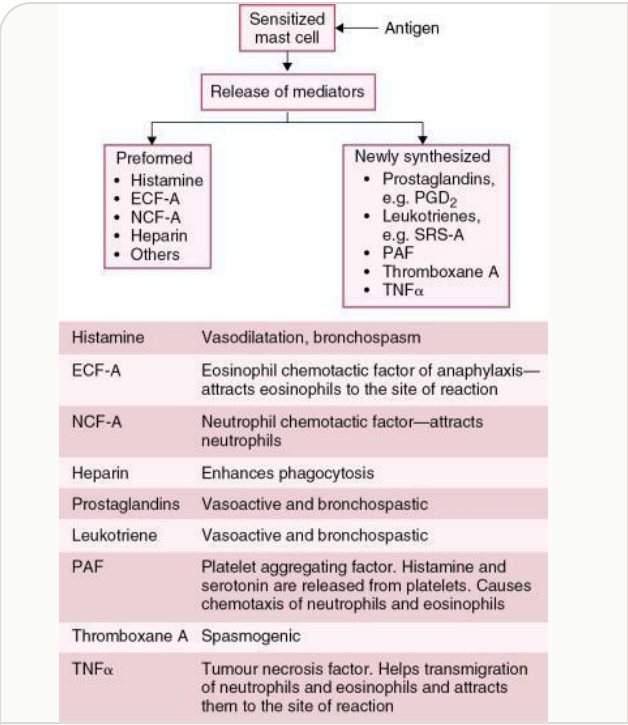
FIGURE 1 (A) The IgE antibody: the Fc end fixes to the mast cell, the Fab end binds allergen. (B) Release of mediators when allergen bridges two IgE molecules.
Dhingra, fig. 30.1

3.3.3 Early and late phases

- **Early phase:** within 5 to 30 minutes: sneezing, rhinorrhoea, blockage, from histamine.
- **Late phase:** 2 to 8 hours later, without further exposure: eosinophils, basophils and T cells infiltrate, causing congestion and thick secretions.

3.3.4 Priming

Mucosa sensitised by one allergen reacts to smaller doses later, and to non-specific stimuli as well.



Histamine	Vasodilatation, bronchospasm
ECF-A	Eosinophil chemotactic factor of anaphylaxis—attracts eosinophils to the site of reaction
NCF-A	Neutrophil chemotactic factor—attracts neutrophils
Heparin	Enhances phagocytosis
Prostaglandins	Vasoactive and bronchospastic
Leukotriene	Vasoactive and bronchospastic
PAF	Platelet aggregating factor. Histamine and serotonin are released from platelets. Causes chemotaxis of neutrophils and eosinophils
Thromboxane A	Spasmogenic
TNFα	Tumour necrosis factor. Helps transmigration of neutrophils and eosinophils and attracts them to the site of reaction

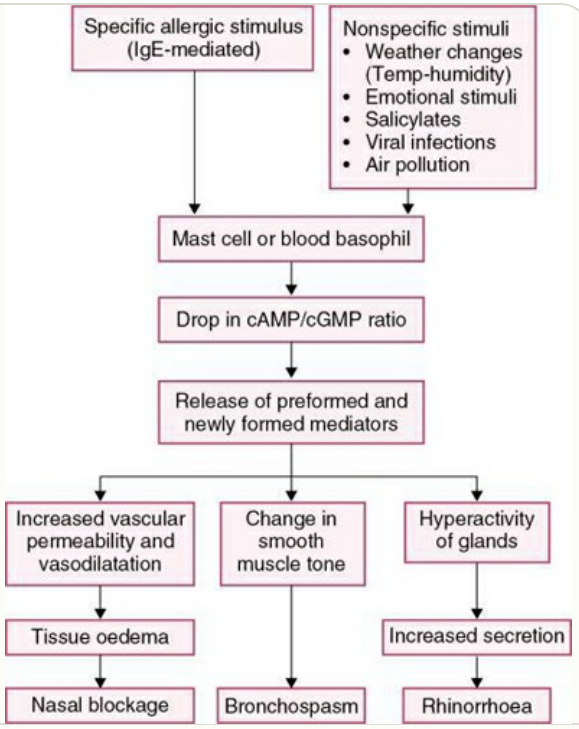


FIGURE 2 First, mediators released from a mast cell. Second, allergic and non-specific stimuli acting through the same mast cells.

Dhingra, figs 30.2 and 30.3

3.4 Clinical features

3.4.1 Symptoms

Usually starting at 12 to 16 years, though any age is possible. Seasonal disease: paroxysmal sneezing (10 to 20 at a time), watery discharge, blockage and itching of the nose, eyes, palate or throat. Perennial disease is milder but persistent: frequent colds, a stuffy nose, reduced smell, postnasal drip, cough and ear symptoms.

3.4.2 Signs

- **Nose:** pale, oedematous, sometimes bluish mucosa; swollen turbinates; watery discharge; a transverse nasal crease from the *allergic salute*.
- **Eyes:** lid oedema, cobblestone conjunctiva, dark circles (*allergic shiners*).
- **Ears:** retracted drum or effusion.
- **Pharynx and larynx:** granular pharyngitis, hoarseness.

- **Children:** the long face and open mouth of chronic mouth breathing.

3.5 ARIA classification

ARIA CLASSIFICATION

CATEGORY	DEFINITION
Intermittent	Symptoms less than 4 days a week, or for less than 4 weeks
Persistent	Symptoms more than 4 days a week and for more than 4 weeks
Mild	None of: disturbed sleep, impaired daily activities or sport, impaired school or work, troublesome symptoms
Moderate to severe	One or more of these

3.6 Investigations

- **Nasal smear:** eosinophils, best taken during active symptoms (also seen in NARES).
- **Blood count:** peripheral eosinophilia, inconsistent.
- **Skin prick test:** allergen pricked into the skin of the forearm or back; a wheal and flare at 10 to 15 minutes is positive, with histamine and diluent as controls.
- **Specific IgE in serum** (formerly RAST): correlates well with skin tests; useful when skin testing is impossible.
- **Nasal provocation test:** reproducing symptoms with allergen.
- **Nasal endoscopy:** to exclude polyps and structural causes.

3.7 Complications

Recurrent sinusitis, nasal polyps (in about 2%), otitis media with effusion, dental and facial growth problems from mouth breathing, and asthma.

3.8 Treatment

3.8.1 Avoidance

Most effective with a single allergen: remove the pet, encase pillows and mattresses, reduce dust mites, change work if necessary.

3.8.2 Drugs

- **Intranasal corticosteroids:** the most effective single treatment for persistent or moderate to severe disease. Teach technique: aim away from the septum.
- **Antihistamines:** oral non-sedating or intranasal, for sneezing, itch and rhinorrhoea; less effect on blockage.
- **Decongestants:** a short course only; oral ones are combined with antihistamines.
- **Sodium cromoglycate:** stabilises mast cells; useful before exposure.
- **Ipratropium bromide:** for watery rhinorrhoea.
- **Leukotriene receptor antagonists:** montelukast, especially with asthma.
- **Oral steroids:** a short course for severe, uncontrolled episodes only.
- **Anti-IgE (omalizumab):** for severe allergic asthma; not a routine treatment for rhinitis.

3.8.3 Step care (ARIA)

1. **Mild intermittent**
   Oral antihistamine or intranasal cromoglycate.
2. **Moderate or persistent**
   Intranasal corticosteroid.
3. **Severe**
   Intranasal steroid with a non-sedating antihistamine.
4. **Uncontrolled**
   Short oral steroid course; consider immunotherapy.

3.8.4 Immunotherapy

Gradually increasing doses of the allergen, by subcutaneous injection or sublingually, until a maintenance dose is reached; it raises specific IgG (blocking antibody) and reduces IgE. Continued for about three years; stopped if no benefit after uninterrupted treatment. The only treatment that alters the course of the disease, for patients whose symptoms are not controlled by drugs or who are sensitised to a single dominant allergen.

PEARLS

- Rhinitis and asthma are one airway: treat both.

- Most steroid-spray failures are technique failures.
- Allergic children with blocked noses may also have big adenoids.

4 CHAPTER FOUR Vasomotor rhinitis

Non-allergic, hyperreactive.

4.1 What it is

A non-allergic rhinitis that mimics allergy, with negative allergy tests, from autonomic imbalance with parasympathetic overactivity and a hyperreactive mucosa. Emotions influence it through the hypothalamus.

4.2 Clinical features

- Paroxysmal sneezing on getting out of bed.
- Profuse, watery rhinorrhoea, sometimes the main complaint; dripping on leaning forward must be distinguished from CSF rhinorrhoea.
- Blockage alternating from side to side, worse at night and on the dependent side.
- Postnasal drip.

The turbinate mucosa is congested and hypertrophic, or normal.

4.3 Complications

Hypertrophic rhinitis, polyps and sinusitis.

4.4 Treatment

4.4.1 Medical

Avoid triggers; antihistamines and oral decongestants; topical steroids; ipratropium for rhinorrhoea; a short systemic steroid course in severe cases; address psychological factors.

4.4.2 Surgical

Reduce the turbinates when obstruction persists; correct a deviated septum or polyps. *Vidian neurectomy*, cutting the parasympathetic secretomotor fibres, for intractable rhinorrhoea; now largely replaced by endoscopic posterior nasal nerve procedures.

5 CHAPTER FIVE Other non-allergic rhinitis

Named causes.

5.1 Drug-induced

Antihypertensives that block sympathetic activity (reserpine, methyldopa, beta-blockers), anticholinesterases, oral contraceptives, and aspirin or NSAIDs in sensitive patients.

5.2 Rhinitis medicamentosa

Rebound congestion from overuse of topical decongestant drops beyond about five to seven days. Treated by stopping the drops, a topical or short systemic steroid course, and turbinate reduction if hypertrophy persists.

5.3 Hormonal

- **Rhinitis of pregnancy:** oedematous mucosa from oestrogen; treated with saline, topical steroids and limited decongestant use; avoid newer antihistamines.
- **Hypothyroidism:** parasympathetic predominance; relieved by thyroxine.
- **Honeymoon rhinitis:** congestion with sexual excitement.

5.4 NARES

Non-allergic rhinitis with eosinophilia syndrome: eosinophils on smear but negative allergy tests; responds to topical steroids; may precede aspirin sensitivity and

polyps.

5.5 Gustatory rhinitis

Watery rhinorrhoea with spicy food, a cholinergic reflex; ipratropium before meals.

5.6 Emotional rhinitis

Anxiety, tension or grief causing rhinitis; counselling.

5.7 Non-airflow rhinitis

After laryngectomy or tracheostomy, when air no longer passes through the nose.

5.8 Local allergic rhinitis

Allergy-like symptoms with negative skin and blood tests but specific IgE produced in the nasal mucosa, shown by nasal provocation; managed like allergic rhinitis.

6 CHAPTER SIX Turbinate surgery

When medicines are not enough.

6.1 Indications

Persistent obstruction from inferior turbinate hypertrophy despite a proper trial of topical steroids, or rhinitis medicamentosa with fixed hypertrophy.

6.2 Methods

Submucosal reduction (microdebrider or radiofrequency), outfracture, partial turbinectomy, laser or cryosurgery. Preserve the mucosa: excessive removal leaves an *empty nose*, with crusting and paradoxical blockage.

OPERATIVE GUIDES IN THEATRE 1 operation

Surgery of the Turbinate
Turbinate reduction after medical care

ALLERGY TESTING, TREATMENT AND THE UNITED AIRWAY (OXFORD TEXTBOOK)

Allergic rhinitis affects about one person in five in Western countries and usually starts around three years of age, after atopic dermatitis and before allergic asthma, the sequence known as the **allergic march**; it is rare before two because sensitisation needs prolonged exposure. Non-allergic rhinitis typically begins in early adulthood and is linked to adult-onset asthma, whereas vasomotor rhinitis is not.

Testing. A skin-prick wheal more than 3 mm larger than the negative control is positive, but only about half of people with positive tests have allergic disease, so tests are interpreted with the history and requested mainly when immunotherapy or targeted avoidance is planned. A small panel (house dust mite, grass, tree and weed pollen, cat, dog and moulds) covers most allergy. Serum specific IgE is a little less sensitive than skin testing because tissue IgE levels are much higher than serum levels.

Treatment. Intranasal steroids take days to start working and about two weeks to reach full effect; modern sprays with low bioavailability are preferred in children. Steroid sprays combine well with oral antihistamines, and a combined steroid–azelastine spray helps patients not controlled by either. Depot steroid injections are not recommended. **Immunotherapy**, subcutaneous or sublingual, is the only disease-modifying treatment and is given for about three years; the sublingual route avoids injections and has an almost negligible anaphylaxis risk. Allergen-impermeable mattress and pillow covers are the best-studied avoidance measure for house dust mite. In turbinate surgery, reducing the **conchal bone** gives the longest-lasting effect.

The united airway. Most patients with allergic asthma have allergic rhinitis, and treating the nose often improves asthma control, so immunotherapy and turbinate reduction can be offered earlier in patients with both conditions.

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Pathogenesis** of allergic rhinitis, and early versus late phase.
- **Signs:** allergic salute, transverse crease, allergic shiners, pale boggy mucosa.
- **ARIA classification** and step care.
- **Skin prick test** technique and controls.
- **Immunotherapy:** mechanism and duration.
- **Vasomotor rhinitis** and the forms of non-allergic rhinitis.
- **Rhinitis medicamentosa.**

7.2 Common pitfalls

COMMON PITFALLS

- Calling one-sided symptoms allergy without examining the nose.
- Prescribing decongestant drops without a time limit.
- Sending a patient for turbinate surgery before a proper trial of steroid spray.
- Missing CSF rhinorrhoea in a patient with clear, one-sided dripping.

Written for this app from Oxford Textbook of Otolaryngology ch. 29, 42, Oxford Specialist Handbook ch. 20, Oxford Handbook of ENT ch. 9, Dhingra ch. 27, 30, 31 and K. J. Lee ch. 27, 31, 59. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

Nasal Polyps

IMAGING FINDINGS

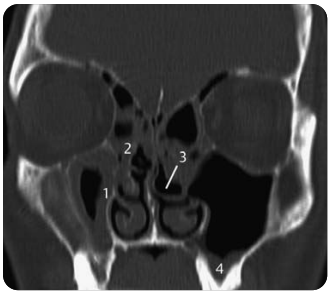
- Polypoid soft tissue filling the nose and all sinuses, both sides
- Widening of the ethmoid labyrinth and thinned septa
- Hyperdense contents suggest fungus or inspissated mucus

TOP DIFFERENTIALS

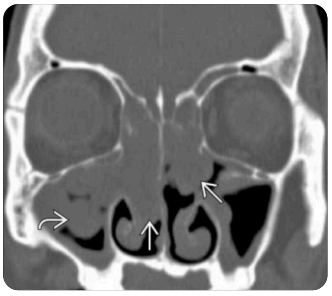
- Allergic fungal sinusitis
- Cystic fibrosis
- Inverted papilloma (unilateral)

CHECKLIST

- Bone erosion or asymmetry: think tumour or fungus
- Pre-operative CLOSE checklist



CT, coronal. The scan shows mucosal thickening in the right maxillary sinus (1), partial opacification of the ethmoid cells (2) on both sides, pneumatizati…



Coronal bone CT shows the classic appearance of sinonasal polyposis

Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 708

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 708.